# Partially inelastic surface — Solution

X22 (2022) — English translation

## A1 — 0.10 pts

*Express $\dot{v}_x$ in terms of $F_x$ and $m$.*

From the theorem on the motion of the center of mass

$$F_x = m\dot{v}_x$$

from which

$$\dot{v}_x = \frac{F_x}{m}$$

**Answer:**

$$\dot{v}_x = \frac{F_x}{m}$$

## A2 — 0.40 pts

*Express $\dot{\omega}$ in terms of $F_x$, $N$, $L$, $\varphi$ and $m$.*

From the equation of dynamics of rotational motion relative to the center of mass

$$I_O\dot{\omega} = \frac{mL^2\dot{\omega}}{3} = NL\sin\varphi - F_xL\cos\varphi$$

from which

$$\dot{\omega} = \frac{3}{mL}\left(N\sin\varphi - F_x\cos\varphi\right)$$

**Answer:**

$$\dot{\omega} = \frac{3}{mL}\left(N\sin\varphi - F_x\cos\varphi\right)$$

## A3 — 0.10 pts

*Express $v_{Ax}$ in terms of $\omega$, $v_x$, $\varphi$ and $L$.*

From the theorem on the addition of velocities

$$\vec{v}_A = \vec{v}_O + \vec{v}_{\mathrm{rel}}$$

from which

$$v_{A_x} = v_x - \omega L\cos\varphi$$

**Answer:**

$$v_{A_x} = v_x - \omega L\cos\varphi$$

## A4 — 0.40 pts

*Prove that $v_{Ax} \le 0$ for any values of $\mu$ and $\varphi$.*

If $v_{A_x} > 0$, then $F_x = -\mu N$ Then

$$\omega L = -\frac{3v_x}{\mu}\left(\sin\varphi + \mu\cos\varphi\right)$$

or same

$$v_{A_x} = v_x \left(1 + \frac{3\cos\varphi}{\mu}\left(\sin\varphi + \mu\cos\varphi\right)\right)$$

But at the same time $v_x < 0$, therefore $v_{A_x} < 0$, which leads to a contradiction.

## A5 — 0.10 pts

*Express $v_{A_x}$ in terms of $v_x$, $\varphi$ and $\mu$.*

In this case $F_x = \mu N$. Then

$$\omega L = \frac{3v_x}{\mu}\left(\sin\varphi - \mu\cos\varphi\right)$$

or same

$$v_{A_x} = v_x \left(1 - \frac{3\cos\varphi}{\mu}\left(\sin\varphi - \mu\cos\varphi\right)\right)$$

**Answer:**

$$v_{A_x} = v_x \left(1 - \frac{3\cos\varphi}{\mu}\left(\sin\varphi - \mu\cos\varphi\right)\right)$$

## A6 — 0.60 pts

*At what minimum possible value $\mu_0$ is slippage impossible for any values of angle $\varphi$?*

Considering that

$$\sin 2\varphi = 2\sin\varphi\cos\varphi$$

$$\cos^2\varphi = \frac{1+\cos 2\varphi}{2}$$

after simple transformation

$$v_{A_x} = -\frac{v_x}{2\mu}\left(3\sin 2\varphi - 3\mu\cos 2\varphi - 5\mu\right)$$

Introducing variable $\varphi_0 = \arctan\mu$

$$v_{A_x} = -\frac{v_x}{2\mu}\left(3\sqrt{1+\mu^2}\sin(2\varphi - \varphi_0) - 5\mu\right)$$

Condition $v_{A_x} < 0$ is realized then when expression in bracket positive. Since $\sin(2\varphi - \varphi_0) \le 1$

$$\frac{5\mu}{3\sqrt{1+\mu^2}} \le 1$$

or same

$$\mu_0 = \frac{3}{4}$$

**second method**

For presence slipping $F_x = \mu N$. Then can express $\mu$ through $\varphi$ as

$$\mu = \frac{3\cos\varphi\sin\varphi}{1+3\cos^2\varphi}$$

Find derivative

$$\frac{d\mu}{d\varphi} = \frac{3}{1+3\cos^2\varphi}\left((\cos^2\varphi - \sin^2\varphi)(1+3\cos^2\varphi) + 6\sin^2\varphi\cos^2\varphi\right)$$

from which find point of the extremum

$$\varphi = \arctan(2)$$

and correspond maximum value

$$\mu_0 = \frac{3}{4}$$

**Answer:**

$$\mu_0 = \frac{3}{4}$$

## A7 — 0.80 pts

*For $\mu < \mu_0$, find all values of the angle $\varphi$ at which slippage is possible. Express the boundaries of the range of angles using $\mu$.*

The condition for the positive value in parentheses is

$$\arcsin\frac{5\mu}{3\sqrt{1+\mu^2}} \leq 2\varphi - \varphi_0 \leq \pi - \arcsin\frac{5\mu}{3\sqrt{1+\mu^2}}$$

from which

$$\frac{\arcsin\frac{5\mu}{3\sqrt{1+\mu^2}} + \arctan\mu}{2} \leq \varphi \leq \frac{\pi - \arcsin\frac{5\mu}{3\sqrt{1+\mu^2}} + \arctan\mu}{2}$$

**second method**

Solve equation

$$\mu = \frac{3\cos\varphi\sin\varphi}{1+3\cos^2\varphi}$$

as a result transformation which obtain two root

$$\varphi_1 = \arctan\frac{3-\sqrt{9-16\mu^2}}{2\mu}, \quad \varphi_2 = \arctan\frac{3+\sqrt{9-16\mu^2}}{2\mu}$$

from which

$$\arctan\frac{3-\sqrt{9-16\mu^2}}{2\mu} \leq \varphi \leq \arctan\frac{3+\sqrt{9-16\mu^2}}{2\mu}$$

which is identical to the answer obtained by the first method.

**Answer:**

$$\arctan\frac{3-\sqrt{9-16\mu^2}}{2\mu} \leq \varphi \leq \arctan\frac{3+\sqrt{9-16\mu^2}}{2\mu}$$

## B1 — 0.20 pts

*Express $v_x$ in terms of $\omega$, $L$ and $\varphi$.*

In this case, the speed of point $A$ is directed vertically, so

$$v_x = \omega L\cos\varphi$$

**Answer:**

$$v_x = \omega L\cos\varphi$$

## B2 — 0.60 pts

*Express $v_y$ in terms of $v_0$, $\omega$, $L$ and $\varphi$.*

The angular momentum of the rod relative to point $A$ upon impact is conserved. The expression for it is

$$\vec{L}_A = \vec{L}_O + m[\vec{AO}\times\vec{v}_O] = \frac{mL^2\vec{\omega}}{3} + m[\vec{AO}\times\vec{v}_O]$$

Thus

$$mLv_0\sin\varphi = mLv_y\sin\varphi + mLv_x\cos\varphi + \frac{mL^2\omega}{3}$$

Account for relation $v_x$ and $\omega$, find in previous part

$$v_y = v_0 - \frac{\omega L(1+3\cos^2\varphi)}{3\sin\varphi}$$

**Answer:**

$$v_y = v_0 - \frac{\omega L(1+3\cos^2\varphi)}{3\sin\varphi}$$

## B3 0.80 pts

*Find the maximum energy $W$ spent on surface deformation. Express your answer in terms of $m$, $v_0$ and $\varphi$.*

For $v_{Ay}$ we have

$$v_{Ay} = v_y - \omega L \sin\varphi = v_0 - \frac{4\omega L}{3\sin\varphi}$$

For energy deformation surface $E_N$ have

$$E_N = \int N v_{Ay} dt$$

But from theorem o motion of the center of mass and part $B2$ have

$$N = -m\dot{v}_y = \frac{m\dot{\omega}L(1+3\cos^2\varphi)}{3\sin\varphi}$$

Then

$$E_N(\omega) = \int_0^{\omega} \frac{m(1+3\cos^2\varphi)}{3\sin\varphi}\left(v_0 - \frac{4\omega L}{3\sin\varphi}\right) L\dot{\omega}dt = \frac{m(1+3\cos^2\varphi)}{3\sin\varphi}\left(v_0\omega L - \frac{2\omega^2 L^2}{3\sin\varphi}\right)$$

Maximum value $W$ be achieved for $v_{A_y} = 0$, thus in this moment 13

**Answer:**

$$W = \frac{mv_0^2(1+3\cos^2\varphi)}{8}$$

## B4 1.20 pts

*Find $\omega_{to}$ after the hit. Express your answer in terms of $v_0$, $L$, $\varphi$ and $k$.*

At the end of the strike

$$E_N = E_{\text{def}} = W(1-k) = \frac{mv_0^2(1+3\cos^2\varphi)(1-k)}{8}$$

from which

$$\frac{m(1+3\cos^2\varphi)}{3\sin\varphi}\left(v_0\omega L - \frac{2\omega^2L^2}{3\sin\varphi}\right) = \frac{mv_0^2(1+3\cos^2\varphi)(1-k)}{8}$$

or

$$\omega^2L^2 - \frac{3v_0\omega L\sin\varphi}{2} + \frac{9v_0^2\sin^2\varphi(1-k)}{16}$$

Solve square equation

$$\omega = \frac{3v_0\sin\varphi(1\pm\sqrt{k})}{4L}$$

Necessarily choose larger from root. Thus

$$\omega_{\text{to}} = \frac{3v_0\sin\varphi(1+\sqrt{k})}{4L}$$

**Answer:**

$$\omega_{\text{to}} = \frac{3v_0\sin\varphi(1+\sqrt{k})}{4L}$$

## B5 0.30 pts

*Find the velocity of point $A$ of rod $v_{Ato}$ immediately after the impact. Express your answer using $v_0$, $\varphi$ and $k$.*

$$v_{Ay} = v_0 - \frac{4\omega_{\text{to}} L}{3\sin\varphi} = -v_0\sqrt{k}$$

or

$$v_{A\text{to}} = v_0\sqrt{k}$$

**Answer:**

$$v_{A\text{to}} = v_0\sqrt{k}$$

## B6 — 0.40 pts

*At what distance $L_{AC}$ from point $A$ is point $C$ of the rod, the speed of which is minimal immediately after the impact? Express your answer using $L$, $\varphi$ and $k$.*

Let's find a point that does not have a velocity component, directed perpendicular to the rod, its speed will be minimal

$$L_{AC} = \frac{v_{A\text{to}}\sin\varphi}{\omega_{\text{to}}} = L\frac{4\sqrt{k}}{3(1+\sqrt{k})}$$

**Answer:**

$$L_{AC} = L\frac{4\sqrt{k}}{3(1+\sqrt{k})}$$

## C1 — 0.40 pts

*Express $\omega$ and $v_y$ in terms of $v_x$, $v_0$, $L$, $\mu$ and $\varphi$.*

$\omega$ was found earlier

$$\omega = \frac{3v_x}{\mu L}\left(\sin\varphi - \mu\cos\varphi\right)$$

From theorem o motion of the center of mass

$$m\dot{v}_y = -N$$

from which

$$v_y = v_0 - \frac{v_x}{\mu}$$

**Answer:**

$$\omega = \frac{3v_x}{\mu L}\left(\sin\varphi - \mu\cos\varphi\right), v_y = v_0 - \frac{v_x}{\mu}$$

## C2 — 1.20 pts

*Find the maximum energy $W$ spent on surface deformation. Express your answer in terms of $m$, $v_0$, $\mu$ and $\varphi$.*

For $v_{Ay}$ we have

$$v_{Ay} = v_y - \omega L\sin\varphi = v_0 - \frac{v_x}{\mu} - \frac{3v_x\sin\varphi}{\mu}\left(\sin\varphi - \mu\cos\varphi\right)$$

and also

$$N = \frac{m\dot{v}_x}{\mu}$$

from which

$$E_N = \int N v_{Ay} dt = \int_0^{v_x} \frac{m}{\mu^2}(\mu v_0 - v_x(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))dv_x$$

or

$$E_N = \frac{m}{\mu^2}\left(\mu v_0 v_x - \frac{v_x^2(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi)}{2}\right)$$

Maximum value $W$ be achieved for $v_{A_y} = 0$, thus in this moment

$$v_x = \frac{\mu v_0}{1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi)}$$

from which

$$W = \frac{mv_0^2}{2(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))}$$

**Answer:**

$$W = \frac{mv_0^2}{2(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))}$$

## C3 — 1.40 pts

*Find $v_{xto}$ after the hit. Express your answer in terms of $v_0$, $\mu$, $\varphi$ and $k$.*

At the end of the strike

$$E_N = E_{\text{def}} = W(1-k) = \frac{mv_0^2(1-k)}{2(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))}$$

from which

$$\frac{m}{\mu^2}\left(\mu v_0 v_x - \frac{v_x^2(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi)}{2}\right) = \frac{mv_0^2(1-k)}{2(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))}$$

or

$$v_x^2 - \frac{2\mu v_0 v_x}{1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi)} + \frac{\mu^2 v_0^2(1-k)}{(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))^2} = 0$$

Solve square equation

$$v_x = \frac{\mu v_0(1 \pm \sqrt{k})}{(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))}$$

Necessarily choose larger from root. Thus

$$v_{x\text{to}} = \frac{\mu v_0(1 + \sqrt{k})}{(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))}$$

**Answer:**

$$v_{x\text{to}} = \frac{\mu v_0(1 + \sqrt{k})}{(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))}$$

## C4 — 1.00 pts

*Find the work done by the friction force $A_{fr}$ during the impact. Express your answer in terms of $m$, $v_0$, $\mu$, $\varphi$ and $k$.*

For the work of the friction force we have

$$A_{\text{fr}} = \int F_x v_{Ax} dt = \int_0^{v_x} mv_x\left(1 - \frac{3\cos\varphi}{\mu}(\sin\varphi - \mu\cos\varphi)\right) dv_x$$

or

$$A_{\text{fr}} = \frac{mv_x^2}{2}\left(1 - \frac{3\cos\varphi}{\mu}(\sin\varphi - \mu\cos\varphi)\right)$$

Substitute find value $v_x$

$$A_{\text{fr}} = \frac{\mu m v_0^2(1 + \sqrt{k})^2(\mu - 3\cos\varphi(\sin\varphi - \mu\cos\varphi))}{2(1 + 3\sin\varphi(\sin\varphi - \mu\cos\varphi))^2}$$

**Answer:**

$$A_{\text{fr}} = \frac{\mu m v_0^2 (1+\sqrt{k})^2 (\mu - 3\cos\varphi(\sin\varphi - \mu\cos\varphi))}{2(1+3\sin\varphi(\sin\varphi - \mu\cos\varphi))^2}$$